(3) $L(s) = \frac{1}{s^5 + 2s^4 + 5s^3 + 10s^2 + 7s + 12}$

a) The first condition requires that the coefficient of each part of the denominator be the same sign. Since, all coefficients are positive, the first confition is met.

b) The second condition requires that no coefficients vanish (equal zero). Since, all coefficients of s^5 through the constant are present, the second condition is met.

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d) There were **2 sign changes** as the third coefficient is negative and all others are positive.

e) Because there are 2 sign changes, this means there are **2 poles** in the right hand plane.

f) The 2 right hand poles ~~are~~ are causing the system to be **unstable**. The conditions are that all poles must be on the left hand plane for the system to be stable.

$0 + s^5$	1	5	7	0
$+ s^4$	2	10	12	0
$+ s^3$	$\frac{(2\cdot5)-(1\cdot10)}{2}$	$\frac{2\cdot7-1\cdot12}{2}$	$\frac{2\cdot0-1\cdot0}{2}$	0
	$=\frac{0}{2}=0$	$=1$	$=0$	
$- s^2$	$\frac{4\cdot10-2\cdot1}{4}$	$\frac{4\cdot12-2\cdot0}{4}$	$\frac{4\cdot0-2\cdot0}{4}$	0
	$=\frac{40-2}{4}$	$=\frac{48}{4}=12$	$=0$	
$+ s$	$\frac{\frac{40-2}{4}\cdot1-4\cdot12}{\frac{40-2}{4}}$	$\frac{\frac{40-2}{4}\cdot0-4\cdot0}{\frac{40-2}{4}}$	0	0
	$=\frac{40-12\cdot4-2}{40-2}$	$=0$		
$+ s^0$	$12\left(\frac{-12\cdot4+40-2}{40-2}\right)-0\frac{\frac{40-2}{4}}{\left(\frac{-12\cdot4+40-2}{40-2}\right)}=12$			

I feel
I got unlucky

$\lim_{4\to0} 4 = 0$ (+)

$\lim_{4\to0} \frac{40-2}{4} = \frac{-2}{0} = -\infty$ (−)

$\lim_{4\to0} \frac{-12\cdot4^2+40-2}{40-2} = \frac{0+0-2}{0-2} = 1$